BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 200 · Graphs and graph representations

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Where is a value 8 inserted if BST root 10 has no left child?

Q2. What is the purpose of a next pointer in a linked list?

Q3. What does zero-based indexing mean?

Q4. In a BST root 20, left 10, right 30, where does 25 go?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

vertex — A node in a graph.

edge — Connection between vertices.

adjacency list — Representation listing neighbours of each vertex.

adjacency matrix — Two-dimensional matrix marking which vertices are connected.

sparse graph — A graph with relatively few edges compared with possible connections.

New learning

Identify vertices — A, B, C and D are graph vertices.

Follow edges — Edges connect A–B, A–C and B–D. Because this graph is undirected, each connection works both ways.

Write adjacency lists — A:B,C; B:A,D; C:A; D:B. Every undirected edge appears in both endpoint lists.

Worked example

Build adjacency lists for triangle A–B, B–C, A–C.

1. Predict — Every vertex connects to the other two vertices.

2. Confirm and explain — A:B,C; B:A,C; C:A,B. Each undirected edge is listed from both endpoints.

We Do · Practise with guidance

Build adjacency lists for chain A–B–C–D.

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Edges: AB, BC, CD
Adjacency lists: ?
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Predict

A is adjacent to B; D is adjacent to C.

Confirm and explain

Full lists: A:B; B:A,C; C:B,D; D:C.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name the vertices and number of edges in A–B, A–C, C–D.

Question 2

Give adjacency lists for an undirected graph with edges AB, BC, CD.

Question 3

Compare adjacency matrix and adjacency list storage on a sparse graph with many vertices.

Exit Ticket · Answer before checking

In an undirected graph, A–B is represented in both A's and B's adjacency lists.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. As the left child of 10.

Q2. It identifies the next node.

Q3. The first element is at index 0.

Q4. Left child of 30.

You Do pages · Independent questions

Q1. Vertices A,B,C,D; three edges.

Q2. A:B; B:A,C; C:B,D; D:C.

Q3. An adjacency matrix uses $O(V^2)$ space irrespective of edge count; adjacency lists use O(V+E) space and usually save memory for sparse graphs.

Exit Ticket · Model answer

True. Each undirected connection is recorded from both endpoints.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: